AI-BASED EARLY MENTAL HEALTH BREAKDOWN DETECTION FROM SPEECH PATTERNS

MileStone-4 Report
Presented By : Group -6

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Project Overview

GOAL: DETECT EARLY SIGNS OF MENTAL HEALTH BREAKDOWN USING SPEECH PATTERNS

APPROACH: MACHINE LEARNING ON MULTIMODAL DATASETS

FOCUS AREAS:

- EMOTION RECOGNITION
- DEPRESSION DETECTION

KEY IDEA: SPEECH SIGNALS REVEAL PSYCHOLOGICAL STATES

Datasets Used

- RAVDESS → 8-class emotion recognition (acted, controlled)
- DAIC-WOZ → Depression detection + PHQ-8 severity (clinical interviews)
- MODMA → MDD vs. Healthy Control (clinical audio)
- SWELL/WESAD → Physiological stress detection (HRV + EDA signals)

Multiple datasets ensure the pipeline generalizes across both affective and clinical domains

RAVDESS DATASET

- 1,440 AUDIO SAMPLES · 24 ACTORS (12M / 12F)
- 8 EMOTIONS:
  - NEUTRAL, CALM, HAPPY, SAD
  - ANGRY, FEARFUL, DISGUST, SURPRISED
- BALANCED DATASET (~105–112 SAMPLES PER CLASS)
- FEATURES: 338 PER FILE
  - MFCCS (1–40) + DELTAS · CHROMA · MEL SPECTROGRAM
  - SPECTRAL CENTROID/BANDWIDTH/ROLLOFF/CONTRAST
  - ZERO CROSSING RATE · RMS ENERGY · DURATION
- TASK: SPEAKER-INDEPENDENT EMOTION RECOGNITION

RAVDESS SPLIT STRATEGY

- Train: 840 samples
- Validation: 300
- Test: 300

- Actor-level split (NOT random):
  - Prevents speaker identity leakage
  - Test actors are completely unseen during training
  - Ensures reported F1 reflects real-world generalization
  - Random splits inflate performance by ~8–12% F1

DAIC-WOZ DATASET

- 189 clinical interviews
  Split: 107 train / 35 val / 47 test
- Two prediction targets:
  - Binary classification (PHQ-8 ≥ 10 → Depressed)
  - Regression (PHQ-8 score, continuous 0–24)
    — Minimal (0–4) · Mild (5–9) · Moderate (10–14)
    — Mod-Severe (15–19) · Severe (20–24)
- Real-world clinical dataset
  (Virtual interviewer "Ellie", USC research project)

DAIC-WOZ FEATURES

- Total: 448 features (multimodal)
- Acoustic (323):
  - COVAREP features — F0, NAQ, QOQ, H1H2, MCEP, HMPDM
  - Formant statistics — F1–F5 (mean/std/min/max)
- Linguistic (14):
  - Word rate · silence proportion · first-person pronoun density
  - PHQ keyword density · filler words · lexical diversity
- Visual (111):
  - 14 Facial Action Units · head pose (Tx/Ty/Tz/Rx/Ry/Rz)
  - Gaze vectors
- Multimodal fusion captures what any single modality misses

CLASS IMBALANCE PROBLEM

- Training data:
  - 77 healthy · 30 depressed
  - Imbalance ratio: ~2.57:1
- The accuracy trap:
  - A model predicting "healthy" always → 72% accuracy
  - → Clinically useless (misses every depressed patient)
- Asymmetric clinical cost:
  - False Negative = missed patient → high risk
  - False Positive = unnecessary follow-up → low risk
- Therefore: optimize for minority-class recall, not accuracy

HANDLING IMBALANCE

- scale_pos_weight = 2.567 in XGBoost
  (Minority class errors penalized 2.57× more)

- SMOTE deliberately avoided:
  At N=107, synthetic samples risk unrealistic interpolations in 448-dimensional acoustic space

- Primary metric: AUPRC (not accuracy)
  - Random baseline AUPRC ≈ 0.28
  - Any score above 0.28 = genuine learning
  - Directly measures minority-class detection quality

PREPROCESSING STEPS

- Audio feature extraction
  (librosa — 338 features per file)
- Normalization
  RobustScaler (median + IQR) — robust to clinical outliers
  Fitted on train only; applied to val/test
- Data splitting
  Actor-level (RAVDESS) · Subject-level (MODMA/SWELL/WESAD)
  Official splits (DAIC-WOZ)
- Feature engineering
  Delta MFCCs · statistical aggregation (mean/std/min/max)
  SelectKBest(k=100) for MODMA
- Input noise augmentation (training only)
  Gaussian noise (std=0.01–0.02) added to feature
  embeddings each epoch → acts as regularizer

MODEL TRAINING

- Four model families compared per dataset:
  XGBoost · SVM · MLP · Ensemble

- DAIC-WOZ: Separate modality branches
  - Acoustic branch (323 features)
  - Linguistic branch (14 features)
  - Visual branch (111 features)

- Modular design benefits:
  - Each branch independently tuned
  - System works even if one modality is unavailable
  - Branches combined for final prediction via fusion

MODELS USED

- XGBoost
  - Gradient-boosted trees · histogram split-finding
  - Objectives: multi:softprob / binary:logistic / reg:squarederror
- SVM (RBF kernel)
  - Calibrated with CalibratedClassifierCV (isotonic, cv=5)
  - Required for soft-vote ensemble probability output
- MLP (PyTorch)
  - Base: 338 → 256 → 128 → 64 → output
  - Residual: adds skip connections (Nb7+)
  - BatchNorm + Dropout + AdamW
- Ensemble (Soft Voting)
  - AUPRC-weighted combination of all branches
- All compared across datasets under identical conditions

XGBOOST (BEST MODEL)

- Works well on small, high-dimensional datasets
- Optimal hyperparameters:
  - RAVDESS: depth=3 · lr=0.1 · reg_α=0.1 · reg_λ=1.5
  - DAIC-WOZ: depth=2 · lr=0.05 · min_child_weight=5 (shallower trees = better generalization at N=107)
- Key regularization levers:
  - L1/L2 on leaf weights (reg_alpha, reg_lambda)
  - min_child_weight=5 (minimum 5 samples per leaf)
  - subsample=0.7–0.8 · colsample_bytree=0.7–0.8
- Provides SHAP feature importance for interpretability
- Strongest overall performer across all datasets

MLP (NEURAL NETWORK)

- Base architecture:
  - 338 → 256 → 128 → 64 → output
  - Each layer: Linear + BatchNorm + ReLU + Dropout(0.3)
- Residual architecture:
  - Adds skip connections to address vanishing gradients (val F1 dropped 0.46→0.32 in deep nets without skip)
- Training config:
  - AdamW · lr=1e-3 · weight_decay=1e-4
  - Gradient clipping: max_norm=1.0
  - Label smoothing: 0.05 (from Nb7+)
  - Batch size: 64 (RAVDESS)
- Limitation: needs more data than available to beat XGBoost

ENSEMBLE STRATEGY

- Stacking → FAILED
  - Meta-learner (Logistic Regression) had only 35 val
  - samples to learn from — overfits branch biases
  - Result: F1 dropped from 0.602 → 0.381
- Soft Voting (AUPRC-weighted) → SUCCESS
  - Final_prob = Σ(AUPRC_i × prob_i) / Σ(AUPRC_i)
  - No fitting required — principled information-theoretic weighting
  - Result: F1 improved from 0.381 → 0.461 (+0.08)

TRAINING SETUP

- Optimizer: AdamW (lr=1e-3, weight_decay=1e-4)
  Selected from grid: SGD · Adam · AdamW · RMSprop
- Dropout: 0.3 (hidden layers) · 0.2 (final hidden)
- LR Scheduler: CosineAnnealingLR (T_max=100)
  Outperforms StepLR and ReduceLROnPlateau
- Gradient clipping: max_norm=1.0 (from Nb5 onwards)
- Early stopping: patience=10–20 on val macro-F1
- Label smoothing: 0.05 (from Nb7 onwards)
- Seed: 42 (all frameworks — fully reproducible)

EVALUATION METRICS

- Macro-F1 (primary)
- Accuracy
- AUPRC (DAIC)
- RMSE (regression)
- Handles imbalance effectively

Why Macro-F1 over Accuracy:

- Weights all classes equally regardless of support
- Critical for imbalanced sets (DAIC, MODMA)

RESULTS SUMMARY

Dataset	Best Model	Accuracy	Macro-F1	Notes
RAVDESS	XGBoost (depth=3)	0.513	0.499	8-class
DAIC-WOZ	AUPRC Fusion	0.532	0.461	Binary
DAIC-WOZ	XGB Regressor	—	—	RMSE=6.30
MODMA	SVM (C=100, k=100)	0.625	0.619	Binary
MODMA	LOSO cross-val	—	0.731	Mean (std=0.444)
SWELL-HRV	MLP [128,64]	0.783	0.771	3-class stress

Consistent performance across 4 datasets and 3 task types

KEY OBSERVATIONS

- Well-classified emotions (F1 > 0.60):
  - Angry · Fearful · Disgust
  - → Strong, distinctive spectral signatures
- Poorly-classified emotions (F1 < 0.40):
  - Calm ↔ Neutral (bidirectional confusion, flat prosody)
  - Happy ↔ Surprised (shared high-energy profile)
- Feature selection is not optional:
  - MODMA: 500+ features → k=100 gave +0.048 F1
  - More features at same N = worse, not better
- Small data → simpler models win:
  - DAIC N=107 → XGBoost beats MLP in every experiment
- LOSO std=0.444 on MODMA → subject heterogeneity dominates, not model instability

LIMITATIONS

- Small datasets
- No temporal modeling
- High subject variability
- Feature-based (not sequence-based)

CONCLUSION AND FUTURE SCOPE

- Built multi-task mental health detection system
- XGBoost best for small data
- Key improvements:
  - Soft voting
  - Feature selection

Future:

- Transformer / LSTM models
- Temporal speech modeling
- Cross-dataset pretraining

THANK YOU